Coherence Is Not Accuracy: Evaluating Reconciliation, Calibration, and Reliability-Aware View Selection in Tabular Foundation Models

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ABSTRACT

When a tabular foundation model is queried in different orders, its marginal and conditional predictions may imply different joint distributions. Reconciliation can remove those contradictions, but it may also distort the probabilities assigned to observed outcomes. We study this problem using CoRe-TFM, which compares direct chain-rule forecasts, pooling, hard and soft marginal reconciliation, and selection using held-out labels. Controlled experiments demonstrate that the benefit of projection depends on the reliability of the imposed marginals. We then examine 160 saved TabICLv2 and TabPFN-3 configurations on ten paired-target datasets, with four randomized held-out splits and training contexts of 128 or 256 observations. At the 256-row context, arithmetic-reference hard projection decreased marginal inconsistency in all 80 comparisons, while test negative log-likelihood increased in 67. Its dataset-balanced mean NLL changes relative to arithmetic pooling were +0.01831 for TabICLv2 and +0.01178 for TabPFN-3. Raw-view-or-pool selection achieved the best descriptive mean NLL for both models, but the gains were small and were not established after exploratory multiple-comparison correction. A geometric-reference sensitivity analysis gave similar results; a conservative gated repair never selected a repair. Six selected configurations reproduced their archived prediction arrays exactly in fresh GPU runs. Raw-source checks reported matching official OpenML MD5 values for all eleven source records, including a later standalone verification of MIC; historical preprocessing identity remains a separate limitation. Our findings distinguish agreement among a model's probability views from predictive reliability and show why reconciliation decisions require evidence beyond structural inconsistency.

1. Introduction

A tabular foundation model may be asked to predict two related categorical outcomes for the same record. The answers can depend on how the question is posed: first predicting A, first predicting B, or asking for one outcome conditional on the other. Ideally these queries describe the same joint probability distribution. In practice, separately estimated views can disagree. The disagreement is easy to detect, but deciding what to do about it is harder: a mathematically consistent replacement is not necessarily a better prediction.

Tabular in-context learning has made this issue timely. Prior-Data Fitted Networks and TabPFN brought inference through pretrained tabular models into practical use (Müller et al., 2022; Hollmann et al., 2023, 2025); subsequent TabPFN releases and TabICLv2 broadened their coverage and performance (Grinsztajn et al., 2025, 2026; Qu et al., 2026). TabArena provides an evolving benchmark context (Erickson et al., 2025), while recent studies of proper scores and uncertainty show that strong task performance does not settle probabilistic reliability (Landsgesell et al., 2026; Johnson et al., 2026; Costa et al., 2026; Tanna et al., 2026).

Let A and B be categorical outcomes associated with features X. Four repeated queries give

$$p(A \mid X), \qquad p(B \mid X), \qquad p(A \mid B, X), \qquad p(B \mid A, X). \tag{1}$$

Using the chain rule in two directions produces

$$J^{B\to A}(a, b \mid x) = p_B(b \mid x)p_{A|B}(a \mid b, x), \tag{2}$$

$$J^{A\to B}(a, b \mid x) = p_A(a \mid x)p_{B|A}(b \mid a, x). \tag{3}$$

If all four views arise from a common joint distribution, the two expressions agree and marginalize to the corresponding direct predictions. Recent work measures precisely these failures of factorization and marginalization consistency in

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TFMs (Klötergens et al., 2026); compatibility of separately specified conditionals is also well established in probability theory (Majid et al., 2025; Muré, 2019).

Coherence does not, by itself, guarantee a better forecast. A projection can remove every algebraic contradiction and still move probability away from the outcomes that occur. Proper scoring rules and the forecast-combination literature provide the relevant test: whether a distribution assigns appropriate probabilities to observations (Brier, 1950; Gneiting and Raftery, 2007; Gneiting et al., 2007; Genest and Zidek, 1986; Ranjan and Gneiting, 2010; Gneiting and Ranjan, 2013). Our question is therefore practical: *when does a reconciliation help, and when is it safer to retain a simple pool?*

CoRe-TFM compares two raw chain-rule factorizations with pooling and KL-based marginal reconciliation. A validation-selected version can decline reconciliation altogether. The experimental design then separates controlled settings, where the true joint is known and reliability can be manipulated, from real-model settings, where selection must rely on a limited number of labels. The latter is important: the most flexible candidate family may offer more potential improvement but also more opportunities to select the wrong candidate (Cawley and Talbot, 2010).

Study contributions.

1. We separate structural compatibility from forecasting quality, with a strict-propriety argument and controlled reliability perturbations showing why the two objectives need not coincide.

2. We place hard and soft marginal reconciliation, two chain-rule orderings, pooling, and validation-based selection in a common evaluation. The projection algorithms themselves are established; our contribution is the controlled comparison of their behavior on tabular foundation-model probability views.

3. We analyze two distinct real-model protocols: a five-seed robustness study and a matched fixed-test experiment comprising 160 model configurations. Keeping them separate enables comparisons without combining incompatible sampling units.

4. We quantify direct changes in marginal consistency, NLL and Brier score under forced repair; examine the Diamonds exception and the non-activating guarded policy; and verify six archived configurations in a fresh GPU runtime with newly recorded checkpoint fingerprints.

2. Related Work

2.1. Tabular foundation models and reliability

The modern TFM line grew from Prior-Data Fitted Networks, which learn an amortized inference procedure from synthetic tasks (Müller et al., 2022), to TabPFN and later large-scale releases (Hollmann et al., 2023, 2025; Grinsztajn et al., 2025, 2026). TabICLv2 develops a separate open and scalable route to tabular in-context learning (Qu et al., 2026). These models enter a tabular-learning literature in which no single inductive bias dominates every dataset: strong tree ensembles and neural architectures remain competitive in different regimes (Gorishniy et al., 2021; Grinsztajn et al., 2022). TabArena formalizes this moving comparison through a maintained benchmark (Erickson et al., 2025).

The present paper is concerned with the predictive distributions produced by these models rather than only their top-1 decisions. ScoringBench evaluates TFMs with proper scoring rules in a predominantly regression-oriented setting (Landsgesell et al., 2026), while recent uncertainty studies report that strong predictive performance does not automatically imply reliable uncertainty estimates (Johnson et al., 2026; Costa et al., 2026). Recent TFM ensembling work reaches a related conclusion: combining accurate models can improve classification while degrading log loss when the combination sharpens probabilities too aggressively (Tanna et al., 2026). The closest diagnostic study to ours measures whether TFMs agree with themselves under factorization and marginalization identities (Klötergens et al., 2026). CoRe-TFM begins from the same defect but asks a different question: whether correcting it improves the predictive distribution.

2.2. Probability pooling, calibration, and knowledge fusion

Combining uncertain information has a long history in statistics and knowledge-based systems. Classical work distinguishes linear, logarithmic, Bayesian, and axiomatic pooling rules (Genest and Zidek, 1986; Heskes, 1997). Forecast-combination studies show that a linear pool can lose calibration even when its components are calibrated, motivating transformed or recalibrated pools (Ranjan and Gneiting, 2010; Gneiting and Ranjan, 2013). These results

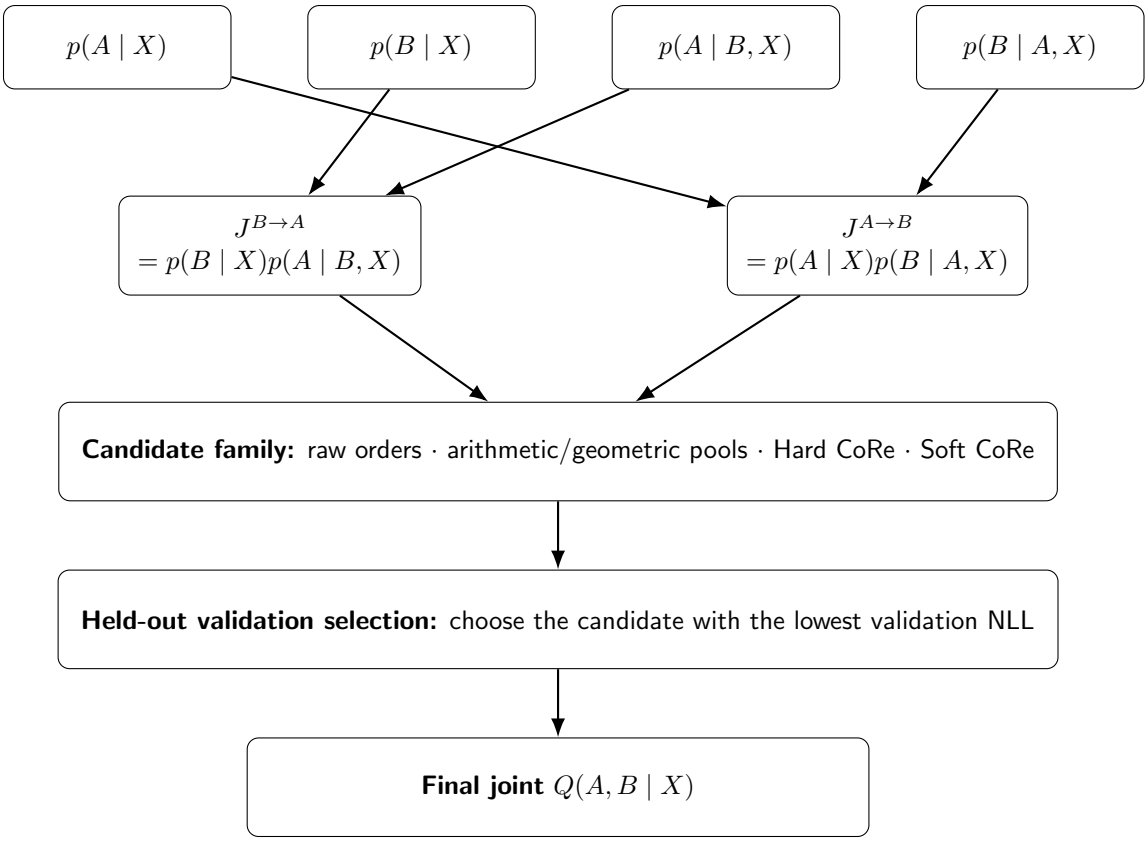


Figure 1: CoRe-TFM workflow. Four overlapping probability views induce two chain-rule joints. Pooling and reconciliation candidates are constructed without retraining the TFM, and a held-out validation set decides whether the final prediction should use a raw order, a pool, or a reconciliation.

are directly relevant to CoRe-TFM because arithmetic and geometric pools are not neutral bookkeeping operations; they change the predictive distribution and must be judged against outcomes.

Calibration provides a second line of related work. Proper scores such as the logarithmic and Brier scores reward probabilistic accuracy rather than only decision accuracy (Brier, 1950; Gneiting and Raftery, 2007). Calibration and sharpness are complementary properties of a useful probabilistic forecast (Gneiting et al., 2007). Post-hoc calibration methods, including temperature scaling, beta calibration, and Dirichlet calibration, use held-out data to correct systematic probability distortions (Guo et al., 2017; Kull et al., 2017, 2019). Recent KBS studies likewise show that post-hoc calibration can be strongly data-dependent (Shaker and Hüllermeier, 2025).

From a knowledge-based-systems perspective, the four TFM outputs are overlapping probabilistic knowledge views. Conflict-aware merging has been studied explicitly as a knowledge-fusion problem (Hunter and Summerton, 2006), and recent KBS work learns data-dependent weights for multi-source decision systems (Zhang and Lin, 2025). CoRe-TFM differs in one important respect: the views are not independent sources. They are direct marginals and conditionals produced by the same frozen model, so the reconciliation objective must account for their statistical overlap rather than treating them as independent experts.

2.3. Compatibility of conditionals

Compatibility of separately specified conditionals is a classical problem. A collection of conditionals may fail to correspond to any common joint distribution, and useful compromises can exist even when exact compatibility is impossible (Muré, 2019). Recent learning-theoretic work studies recovery of a joint from conditional information (Majid et al., 2025). In CoRe-TFM, compatibility is observable through chain-rule and marginalization identities, but

compatibility is not the final target. We evaluate every repaired distribution with held-out proper scores because an internally compatible joint can still be predictively inferior.

2.4. Information projection, matrix scaling, and finite-sample selection

Hard CoRe is an information projection onto the transportation polytope defined by the queried marginals. Its roots include iterative proportional fitting (Deming and Stephan, 1940), I-divergence projection onto convex sets (Csiszár, 1975), and matrix scaling (Sinkhorn, 1964). Modern entropic optimal-transport work provides efficient Bregman and Sinkhorn-style algorithms for related projections (Cuturi, 2013; Benamou et al., 2015; Chizat et al., 2018). Geometric pooling is also connected to logarithmic opinion pools (Heskes, 1997). These projection and scaling procedures are well-established; CoRe-TFM does not claim them as new optimization algorithms.

The methodological novelty lies in deciding whether those primitives should be applied to repeated TFM probability views. Once multiple raw, pooled, and reconciled candidates are available, the problem becomes finite-sample model selection. Validation-based choice can itself overfit when the selection criterion is noisy (Cawley and Talbot, 2010). This motivates the candidate-family ablations and oracle-regret decomposition reported later in the paper. Statistical comparisons across datasets follow the non-parametric paired-testing perspective common in machine-learning benchmark studies (Demšar, 2006).

3. Problem Formulation

Let $A \in \{1, \dots, K_A\}$ and $B \in \{1, \dots, K_B\}$. For an input x, a frozen predictor supplies the four views described above. We define the factorization inconsistency

$$\Delta_{\mathrm{fac}}(x) = \mathrm{TV}\left(J^{B\to A}, J^{A\to B}\right) = \frac{1}{2}\sum_{a,b}\left|J^{B\to A}(a,b \mid x) - J^{A\to B}(a,b \mid x)\right|. \tag{4}$$

The two induced joints preserve different direct marginals automatically: $J^{B\to A}$ preserves p_B and $J^{A\to B}$ preserves p_A. Their opposite marginals may disagree with the direct queries.

Our objective is a normalized joint $Q(A, B \mid X)$ that balances three different notions:

1. *compatibility*: structural agreement among the queried views;
2. *fidelity*: closeness of the post-processed joint to the model's queried distributions; and
3. *predictive quality*: proper-score performance against held-out outcomes.

These notions need not agree. In particular, zero post-repair inconsistency is not evidence that Q is more accurate.

A strict-propriety counterexample. Let R be a positive pooled joint distribution and $Q = \mathcal{P}_{p_A,p_B}(R) \neq R$. Under logarithmic loss, if the true joint distribution is $P^* = R$, the expected score change due to repair is

$$\mathbb{E}_{Y\sim R}[-\log Q(Y)] - \mathbb{E}_{Y\sim R}[-\log R(Y)] = \mathrm{KL}(R\|Q) > 0. \tag{5}$$

Conversely, if $P^* = Q$, the same repair decreases expected log loss by $\mathrm{KL}(Q\|R) > 0$. This is a direct consequence of strict propriety, not a new optimization theorem. It demonstrates why marginal compatibility cannot by itself fix the sign of the predictive effect: the additional assumption needed is that the imposed marginals, and hence the resulting joint, are reliable relative to the unmodified forecast.

The four views are repeated predictions from one frozen model and therefore overlap statistically. They must not be interpreted as independent Bayesian experts. CoRe objectives are regularized compatibility objectives, not likelihood multiplication under conditional independence.

4. CoRe-TFM

4.1. Pooling baselines

For a direction weight $w \in [0, 1]$, the arithmetic pool is

$$Q_{\mathrm{arith},w} = wJ^{B\to A} + (1-w)J^{A\to B}. \tag{6}$$

Table 1
Positioning of CoRe-TFM relative to neighboring lines of work. "Outcome-gated" means that held-out labels can determine whether the transformation is used.

Approach	Multiple probability views	Enforces joint compatibility	Uses held-out labels	Outcome-gated	Primary target
Calibration methods	No	No	Yes	No	Calibration / proper score
Opinion pooling	Yes	No	Optional	No	Consensus forecast
TFM self-consistency diagnostics	Yes	No	No	No	Detect inconsistency
Hard marginal projection	Yes	Yes	No	No	Structural coherence
CoRe-TFM	Yes	Optional	Yes	Yes	Reliability-aware predictive quality

It minimizes $w\,\mathrm{KL}(J^{B\to A}\|Q) + (1-w)\,\mathrm{KL}(J^{A\to B}\|Q)$. The normalized geometric pool is

$$M_w(a,b) = \frac{J^{B\to A}(a,b)^w J^{A\to B}(a,b)^{1-w}}{\sum_{a',b'} J^{B\to A}(a',b')^w J^{A\to B}(a',b')^{1-w}}, \tag{7}$$

which is the corresponding reverse-KL barycenter.

4.2. Hard CoRe

Define the transportation polytope with prescribed direct marginals

$$\mathcal{T}(p_A,p_B) = \left\{ Q \geq 0 \,:\, \sum_b Q(a,b) = p_A(a),\ \sum_a Q(a,b) = p_B(b) \right\}. \tag{8}$$

Let R be a strictly positive reference joint, with the same class support as the four views. The hard marginal reconciliation operator is

$$\mathcal{P}_{p_A,p_B}(R) = \arg\min_{Q\in\mathcal{T}(p_A,p_B)} \mathrm{KL}(Q\|R). \tag{9}$$

Protocol A uses $R = M_w$, the normalized geometric consensus, whereas the independently audited October protocol computes $R = Q_{\mathrm{arith},0.5}$, the arithmetic consensus. These are *different input distributions* and can yield different projected joints; the numerical results of one must not be attributed to the other. The feasible set is nonempty because $p_A p_B$ is feasible. With strictly positive R, the projection is unique and can be obtained by iterative proportional fitting. Hard reconciliation encodes a strong reliability assumption: the two direct marginals should be preserved exactly. Hereafter, "hard repair" in the October audit refers explicitly to $\mathcal{P}_{p_A,p_B}(Q_{\mathrm{arith},0.5})$.

4.3. Soft CoRe

To relax that assumption, Soft CoRe solves

$$\begin{aligned} Q_{\mathrm{soft}} = \arg\min_Q\ & w\,\mathrm{KL}(Q\|J^{B\to A}) + (1-w)\,\mathrm{KL}(Q\|J^{A\to B}) \\ & + \lambda_A\,\mathrm{KL}(Q_A\|p_A) + \lambda_B\,\mathrm{KL}(Q_B\|p_B). \end{aligned} \tag{10}$$

The chain rule for KL divergence yields

$$\mathrm{KL}(Q\|J^{B\to A}) = \mathrm{KL}(Q_B\|p_B) + \mathbb{E}_{b\sim Q_B}\,\mathrm{KL}(Q_{A|B=b}\|p_{A|B=b}), \tag{11}$$

$$\mathrm{KL}(Q\|J^{A\to B}) = \mathrm{KL}(Q_A\|p_A) + \mathbb{E}_{a\sim Q_A}\,\mathrm{KL}(Q_{B|A=a}\|p_{B|A=a}), \tag{12}$$

showing that all four predictive views enter the objective. Generalized scaling uses exponents

$$\tau_A = \frac{\lambda_A}{1+\lambda_A}, \qquad \tau_B = \frac{\lambda_B}{1+\lambda_B}. \tag{13}$$

As $\lambda_A, \lambda_B \to \infty$, the method approaches hard marginal preservation; at small penalties it stays closer to the factorized consensus.

4.4. Selective CoRe

Selective CoRe treats reconciliation as optional. The candidate family includes the two raw orders, equal and weighted arithmetic/geometric pools, weighted Hard CoRe, and weighted Soft CoRe. The implemented grid uses five direction weights

$$\{0, 0.25, 0.5, 0.75, 1\} \tag{14}$$

and seven marginal penalties

$$\{0.03, 0.1, 0.3, 1, 3, 10, 30\}. \tag{15}$$

After duplicate removal, the full family contains 48 candidates. The candidate with the lowest held-out validation NLL is frozen before test evaluation.

For a fixed finite candidate family $\mathcal{C}$ of size M, let all candidates be specified independently of the validation labels, with per-example negative log-likelihood bounded by $B = -\log \epsilon$ after a positive probability floor ϵ. If the validation observations are i.i.d. and L is the corresponding population loss, a union-bound/Hoeffding argument gives

$$L(\hat{Q}) \leq \min_{Q \in \mathcal{C}} L(Q) + 2B\sqrt{\frac{\log(2M/\delta)}{2n_v}}, \tag{16}$$

with probability at least $1 - \delta$. This is a conservative finite-class bound for the selection stage, not a guarantee about the upstream foundation model, non-independent examples, adaptively constructed candidates, or out-of-distribution deployment. The dependence on both M and the validation size n_v motivates the candidate-family and validation-fraction analyses.

4.5. Opportunity and selection regret

To separate method headroom from selection failure, let L_{arith} be arithmetic-pool loss, L_{sel} the selected loss, and L_{oracle} the minimum test loss over the candidate family. We report

$$\text{oracle opportunity} = L_{\text{arith}} - L_{\text{oracle}}, \tag{17}$$

$$\text{selection regret} = L_{\text{sel}} - L_{\text{oracle}}, \tag{18}$$

$$\text{net effect} = L_{\text{sel}} - L_{\text{arith}}. \tag{19}$$

The oracle is analysis-only and is never used to choose a deployable candidate.

5. Experimental Design

5.1. Controlled known-truth studies

We use three controlled layers. First, a multiclass surrogate data-generating process provides exact $P^*(A, B \mid X)$ and therefore exact expected proper scores. Second, an exact-view perturbation benchmark independently corrupts the two direct marginals and the two conditional families, isolating the reliability assumptions encoded by reconciliation. Third, a heterogeneous 100-task mixture varies view reliability from task to task and compares Selective CoRe with fixed methods and a full-family oracle. A separate validation-size study varies the number of labels available to the selector.

5.2. Real-model benchmark

The real-data analysis includes Anneal, Car, Credit, Customer, Diamonds, Marketing, MIC, Nursery, Phishing, and Wine. The Car and Wine benchmarks follow the UCI and wine-quality sources, respectively (Bohanec, 1988; Cortez et al., 2009). Two released TFMs form the primary model comparison: TabICLv2 2.1.1 and TabPFN 8.4.0 using the TabPFN-3 interface. CatBoost 1.2.10 is an explicitly non-TFM boundary baseline and is excluded from estimates labeled "primary TFM."

TabFM was excluded from the frozen experimental protocol for recorded runtime and operational compatibility reasons, not for unfavorable predictive performance. The release required by the adapter was not fully exercised in this

bounded execution environment, and no controlled TabFM timing comparison exists in the evidence archive. It would therefore be inappropriate to infer a slowdown factor or relative accuracy from its absence.

The evaluation uses five outer folds per dataset and model, disjoint training, validation and test partitions, training-set coverage of the target classes, and no more than 128 test observations in a fold. At a 256-row training cap there are 52 validation observations per fold. For one seed the three-model matrix has 10 datasets × 3 models × 5 folds = 150 model–fold tasks and 1,200 reported method rows (eight retained method summaries per task). The 48 candidate distributions inside Selective CoRe should not be confused with those eight reported methods.

5.3. Seed and data-budget robustness

The archived supplementary evidence contains five complete runs at the 256-row cap, using constrained-sampling seeds 11, 23, 42, 71 and 101. Four additional caps (64, 128, 512 and 1024) were evaluated with each of three matched seeds (23, 42 and 71), for a total of 17 runs. The archived aggregate contains 20,400 method-result rows: 17 runs × 10 datasets × 3 models × 5 folds × 8 methods. We used a fixed validation fraction of approximately 20%, giving 13, 26, 52 and 103 validation observations at caps of 64, 128, 256 and 512, respectively. At a nominal cap of 1024, available training data limited the actual cap to 718–1024 observations and validation support to 144–205 (mean 194.4).

The budget experiment is deliberately described as *combined upstream-and-selection evidence sensitivity*: changes to training context and validation support cannot be causally disentangled. The 256-row row uses five seeds unless explicitly marked as the matched-three-seed subset; other caps use three.

5.4. Fixed-test, matched-context evaluation

The matched fixed-test study addresses two limitations of the five-seed study: increasing training context and selector labels together, and comparing candidate families without a strong calibrated-pooling reference. It is a *second, separately identified protocol*; its cells must not be numerically pooled with the historical five-fold/five-seed analysis. The fixed-test archive comprises ten dataset pairs (the same named benchmark collection), two released TFMs (TabICLv2 2.2.0 and TabPFN package 9.2.0 using `ModelVersion.V3`), four randomized held-out split settings per dataset (seed/fold 23/0, 23/1, 42/0, 71/0), and two nested training contexts of 128 and 256 examples. This yields $10 \times 2 \times 4 \times 2 = 160$ real-model prediction caches. The split setting named "fold" is a randomized repeat, not a classical disjoint K-fold partition. The newer model versions and feature-pair handling, including a corrected Diamonds `cut/color` pairing, are recorded as protocol changes. The exact original raw OpenML version and transformed feature schema have not been independently authenticated; cross-protocol numerical merging is therefore prohibited.

For each dataset and split setting, 205 held-out validation examples and 128 held-out test examples are reserved independently of the context cap. Both TFMs and both context sizes use identical validation and test identifiers within the setting; nested context subsets protect the paired comparison. The 205-label validation pool alongside 128 or 256 context examples deliberately *decouples* upstream training context from selection evidence; it is not a 128-total-label learning scenario. This protocol has at least 205 validation labels and 128 test labels in addition to its stated training context. It therefore probes contextual inference and selector sample size rather than ordinary limited-label learning with 128 or 256 total observed labels. We report the protocol-designated central comparison at the 256-example cap and the first 52 validation labels, and use the 128-example cap as a context sensitivity analysis. Per-example NLL, multiclass joint Brier and classification accuracy are scored on the unchanged test set; top-label ECE is descriptive.

Four probabilistic queries are inferred using separately fitted marginal and conditional estimators. The independent audit reconstructs each raw chain-rule joint and the arithmetic/geometric pools, hard marginal projection, partial soft repair, and validation-fitted selectors. The newer eight-base-candidate family is a restricted, explicitly documented counterpart of the original 48-candidate research formulation; results labeled *full-family* below refer to the *new audit's* eight-base-candidate selector, not to the historical 48-way comparison. More precisely, the October 160-cache implementation evaluates *linear partial repair* $Q_\alpha = (1-\alpha)Q_{\text{arith},0.5} + \alpha\mathcal{P}_{p_A,p_B}(Q_{\text{arith},0.5})$ for $alpha \in \{0.25, 0.50, 0.75\}$. This is a convex interpolation and not the penalty-based Soft CoRe optimizer defined earlier. We retain the original Soft CoRe formulation as a historical method, but label the October α-repairs explicitly to avoid implying objective equivalence. The validation-only baselines include raw-or-pool selection, temperature-fitted pooling, and a guarded repair option. Temperature and policy choice use no test labels. For a given split, all candidate losses are scored on the same fixed held-out examples.

For the principal inferential contrasts, we first average four repeated split effects within each dataset separately for each model, then report the unweighted mean over ten datasets. We give paired dataset-resampling bootstrap intervals and exploratory two-sided Wilcoxon tests, followed by Holm adjustment over the documented primary exploratory

policy family. The 160 caches and their 1,920 reconstructed method rows are not interchangeable statistical sample sizes; only ten datasets are treated as independent experimental units for dataset-blocked comparisons. The second-protocol package version, checkpoint identifier, frozen split hash and cache SHA-256 are recorded, but weight bytes loaded during the original benchmark runs and original raw OpenML dataset revisions remain unauthenticated.

5.5. Targeted replication and dataset provenance

After freezing the 160-cache analysis, we performed a targeted reproduction in a fresh Colab GPU runtime using the archived train, validation and test IDs. For each of Car, Nursery and Diamonds, both TabICLv2 and TabPFN-3 were run at training context 256, seed 23 and split index 0, yielding six additional executions of existing configurations. No additional dataset or split was added to the primary $n = 160$ benchmark. The replication compared saved probability tensors and proper scores against the historical caches, recorded the software environment, and fingerprinted the explicitly configured model-file paths. These file hashes document the replication execution; historical cache metadata did not contain authenticated loaded-weight hashes. In a separate online step, we requested the metadata and raw source files for the eleven underlying OpenML records (the Wine pairing joins two source IDs) and compared downloaded MD5 values against the OpenML-reported values. We report inaccessible records rather than substituting another dataset version.

5.6. Outcomes, aggregation and inference

Joint negative log-likelihood (NLL) and joint Brier score are the principal proper scores. Additional diagnostics include conditional NLL, expected calibration error (ECE), total variation between the two factorizations, and distortion of the direct marginals. ECE depends on the selected binning and is descriptive here; known-truth experiments also permit exact expected loss, truth distance and oracle regret.

For a method comparison, we first form the paired score difference within each dataset–model–fold–seed cell. Differences are averaged over folds, over the two TFMs, and then over seeds within each dataset. Thus the confirmatory real-data unit is the *dataset* ($n = 10$), not the individual fold, row, seed, or model cell. A two-sided Wilcoxon signed-rank test is applied to the ten dataset-level paired effects. Because the study contains exploratory choices among methods, budgets and data regimes, the reported p-values should not be interpreted as independently preregistered confirmatory tests. We report model-specific and budget comparisons as exploratory; no family-wise confirmatory inference is claimed for the many additional contrasts (Demšar, 2006). The five-seed 256-row bootstrap interval reported below resamples ten datasets with replacement (20,000 percentile replicates; fixed analysis seed 20261012); it is a sensitivity interval, not a pre-registered equivalence test. Positive "Selective-minus-arithmetic" NLL differences favor arithmetic pooling.

Display precision and practical scale. Displayed losses and differences are rounded for readability. The source CSV/JSON files in S1 retain the unrounded values used in the real-model analyses. An NLL difference also has a direct multiplicative interpretation. If $\Delta = L_{\text{method}} - L_{\text{baseline}}$, then $\exp(-\Delta)$ is the ratio of the geometric-mean probability assigned to the observed class by the method and the baseline. Thus, the controlled Selective-CoRe improvement of about 0.0067 NLL corresponds to roughly a 0.67% increase in geometric-mean probability on the observed class, whereas the five-seed real-data disadvantage of about 0.0009 corresponds to only about a 0.09% decrease. The larger 0.1505-NLL failure of compulsory Hard CoRe corresponds to about a 14.0% decrease. These are multiplicative probability interpretations, not absolute percentage-point changes, and downstream practical value remains application- and decision-cost-dependent.

6. Results

6.1. Exact-view perturbations expose the reliability assumption

Table 3 summarizes the frozen, known-truth perturbation analysis reported in the original controlled-study artifacts. These controlled results are distinguished from the 17-run real-model archive audit; a fresh end-to-end synthetic rerun is not claimed here.

When the direct marginals are reliable, Hard CoRe improves over arithmetic pooling by 0.0133 NLL on average ($p_{\text{Holm}} = 7.11 \times 10^{-15}$). When the conditionals are reliable but direct marginals are noisy, Hard CoRe is worse by 0.1505 ($p_{\text{Holm}} = 7.11 \times 10^{-15}$). Exact marginal preservation therefore transfers the quality of the marginals into the joint; it is not intrinsically beneficial.

Table 2
Two real-model experimental protocols retained as separate evidence streams. The second protocol isolates changes in training context at fixed validation/test IDs; its four randomized split settings are not independent datasets.

Component	Protocol A: five-seed benchmark	Protocol B: fixed-test benchmark	Interpretive consequence
Models	TabICLv2 2.1.1; TabPFN 8.4.0/V3 interface	TabICLv2 2.2.0; TabPFN 9.2.0/V3	Do not pool model-version estimates
Datasets	Ten paired-target datasets	Ten paired-target datasets	Paired identities require feature/preprocessing review
Sampling	Five outer folds; five seeds at 256	Four randomized held-out settings	Different replication units and split construction
Contexts	64–1024 in budget study	128 and 256 with nested context	Fixed test IDs across contexts in Protocol B
Selection labels	Proportional to context (52 at 256)	205 reserved; primary first 52	Selection and context budgets vary separately in Protocol B
Selector family	Up to 48 candidates	Eight base candidates plus calibration policies	Different method-family sizes
Statistical unit	Dataset (ten)	Dataset (ten)	Multiple runs do not create independent datasets

Table 3
Best mean exact expected NLL in each exact-view perturbation regime (50 paired seeds).

Regime	Best method	Expected NLL
Uniform low noise	Geometric pool	1.4786
Direct marginals reliable	Soft CoRe ($\lambda = 10$)	1.4868
Conditionals reliable	Arithmetic pool	1.5088
$B \to A$ reliable	$J^{B\to A}$	1.4753
$A \to B$ reliable	$J^{A\to B}$	1.4753
All four views noisy	Arithmetic pool	1.5387

6.2. Heterogeneous reliability can reward adaptive selection

Table 4 presents the original heterogeneous 100-task controlled benchmark. Its task-level evidence is analyzed separately from the ten-dataset, dataset-blocked real-model comparison.

Selective CoRe improves over arithmetic pooling by 0.0067 mean exact NLL, wins 68/100 tasks, and has a paired Wilcoxon $p = 1.25 \times 10^{-11}$. It selects Soft CoRe 38 times, geometric pooling 33 times, raw orders 19 times, arithmetic pooling 10 times, and Hard CoRe zero times. Its mean regret to the non-deployable full-family oracle is only 0.0007. The controlled validation-size study is consistent with the finite-selection interpretation: mean oracle regret falls from 0.0056 with 100 validation labels to 0.0012 with 800 labels.

6.3. Frozen real benchmark: coherence improves while proper scores worsen

The original frozen 256-row run produced the result in Table 5. This is retained because it was the first prespecified real-model analysis and because the later robustness study changes its interpretation rather than invalidating its execution.

Selective CoRe improves agreement with the direct marginals while worsening both proper scores in this frozen run. The result demonstrates the main conceptual point: a structurally more faithful joint can be predictively worse. Mean factorization TV in the primary TFM cells is nonzero, so the finding is not caused by a trivial absence of incompatibility.

The effect is strongly model dependent. In the frozen run, Selective CoRe is worse than arithmetic by 0.0067 NLL on TabICLv2 but better by 0.0030 on TabPFN-3. The paired difference between model-specific effects is 0.0097 ($p = 0.00977$, exploratory).

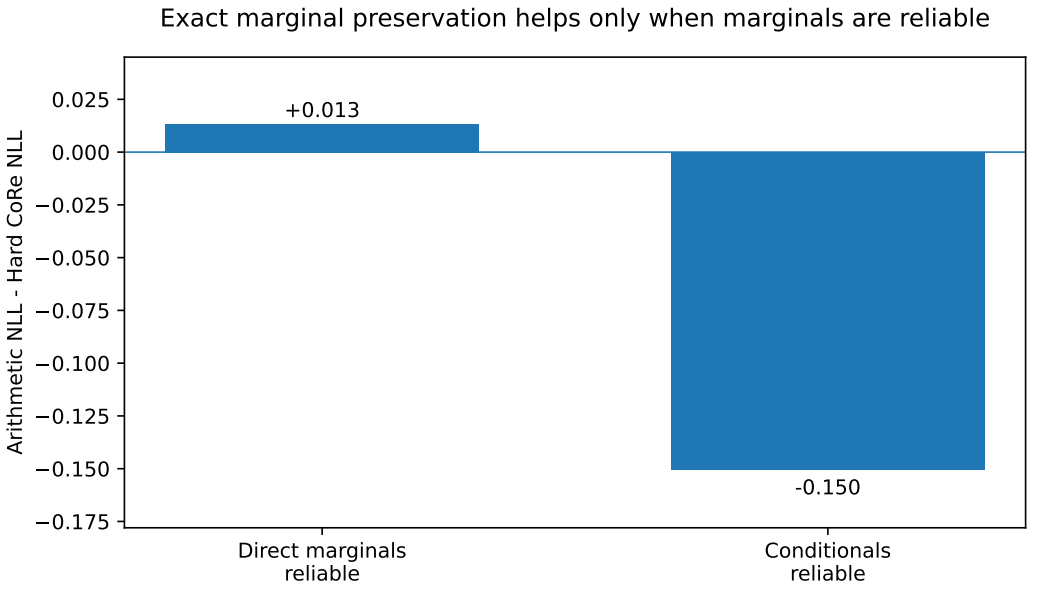


Figure 2: Reliability mechanism in the exact-view perturbation study. The vertical axis is arithmetic NLL minus Hard-CoRe NLL, so positive values favor exact marginal preservation. Hard CoRe helps when direct marginals are reliable and fails sharply when the conditionals are reliable but the direct marginals are noisy.

Table 4
Heterogeneous 100-task controlled benchmark.

Method	Mean NLL	SE	Mean oracle regret
Selective CoRe	**1.4837**	0.0055	**0.0007**
Arithmetic	1.4903	0.0056	0.0074
Geometric	1.4914	0.0056	0.0084
Soft CoRe ($\lambda = 1$)	1.5006	0.0060	0.0176
$J^{A\to B}$	1.5068	0.0070	0.0239
$J^{B\to A}$	1.5144	0.0067	0.0315
Hard CoRe	1.5383	0.0085	0.0554

6.4. Five-seed robustness attenuates the frozen negative result

The completed multi-seed analysis repeats the 256-row protocol for five constrained-sampling seeds. After averaging folds, the two primary TFMs and the five seeds within each dataset, the mean Selective-minus-arithmetic effect is +0.000851 NLL. Selective CoRe is better on two datasets and worse on eight. The two-sided dataset-blocked Wilcoxon result is $p = 0.10547$. An exploratory percentile bootstrap over the ten dataset means gives a 95% interval of approximately [−0.00157, +0.00291] (20,000 replicates, fixed seed 20261012). This interval includes both small benefits and small harms and should not be read as establishing equivalence. The five seed-level mean effects are +0.0004, +0.0009, +0.0018, +0.0009, and +0.0002 for seeds 11, 23, 42, 71, and 101, respectively.

This robustness result does not establish equivalence between Selective CoRe and arithmetic pooling. It does show that the magnitude and inferential strength of the original single-seed disadvantage were sensitive to the sampling realization. The defensible 256-row conclusion is therefore that Selective CoRe has a small average disadvantage under this bounded protocol, with substantial data-set and model heterogeneity and without dataset-blocked significance over the five-seed aggregate.

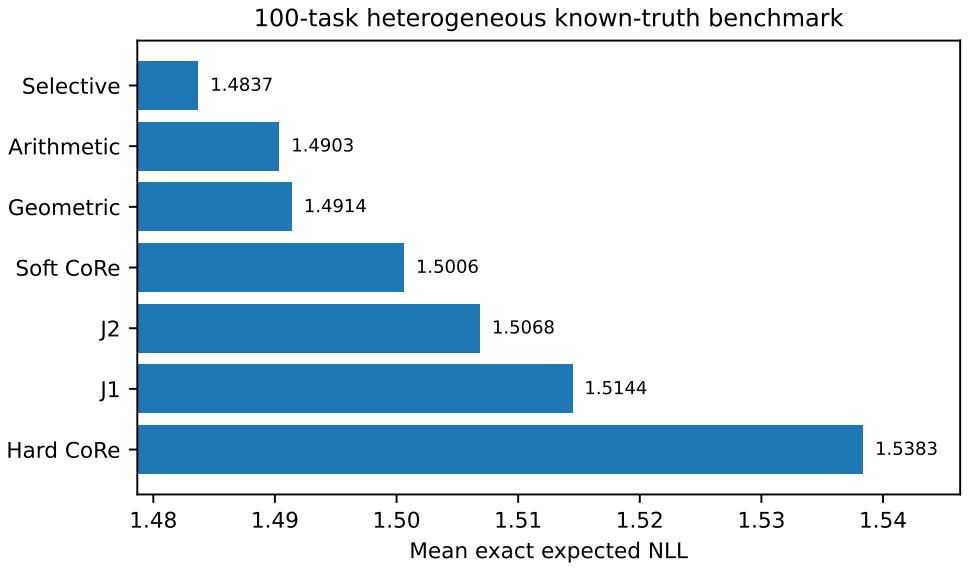


Figure 3: Mean exact expected NLL in the 100-task heterogeneous known-truth benchmark. Selective CoRe has the lowest mean NLL and mean oracle regret, while compulsory Hard CoRe is the worst fixed strategy in this deliberately heterogeneous reliability mixture.

Table 5
Frozen single-seed dataset-blocked Selective-CoRe effects relative to arithmetic pooling across the two primary TFMs ($n = 10$ datasets). Positive proper-score differences are worse.

Metric	Mean difference	95% bootstrap CI	Wins	p
Joint NLL	+0.0018	[+0.0005, +0.0033]	2/10	0.02734
Joint Brier	+0.0006	—	1/10	0.00586
Marginal distortion	−0.0039	—	9/10	0.00391

6.5. Data-budget sensitivity reveals a sign transition

Table 6 summarizes the primary TFM effect over the available data budgets. For the 64, 128, 512, and 1024 caps, three matched seeds are available; the 256 row contains five seeds. A matched-three-seed calculation gives 0.0012 rather than 0.0009 at 256 and leaves the qualitative sequence unchanged.

These are descriptive comparisons, not five separate confirmatory claims of improvement. The experiment contains ten dataset units, some datasets cannot supply the nominal 1024 training rows, and each larger context also supplies more held-out validation examples. The negative difference at the largest cap is therefore evidence of a budget-dependent association in this benchmark, not a causal demonstration that either training context or validation size alone makes reconciliation profitable.

The model-specific pattern strengthens the reliability interpretation (Table 7). TabPFN-3 is near parity at 64 rows and favors Selective CoRe at every larger budget in the aggregate. TabICLv2 begins with a substantially larger selector penalty and crosses to a small selector advantage only at 512 rows.

6.6. Inconsistency is not a universal reconciliation-gain signal

Factorization TV decreases as the evidence budget grows for both TFMs. For TabICLv2, mean TV falls from 0.1237 at the 64-row cap to 0.0560 at 1024; for TabPFN-3 it falls from 0.0601 to 0.0403. Direct and conditional NLLs also improve with increasing budget.

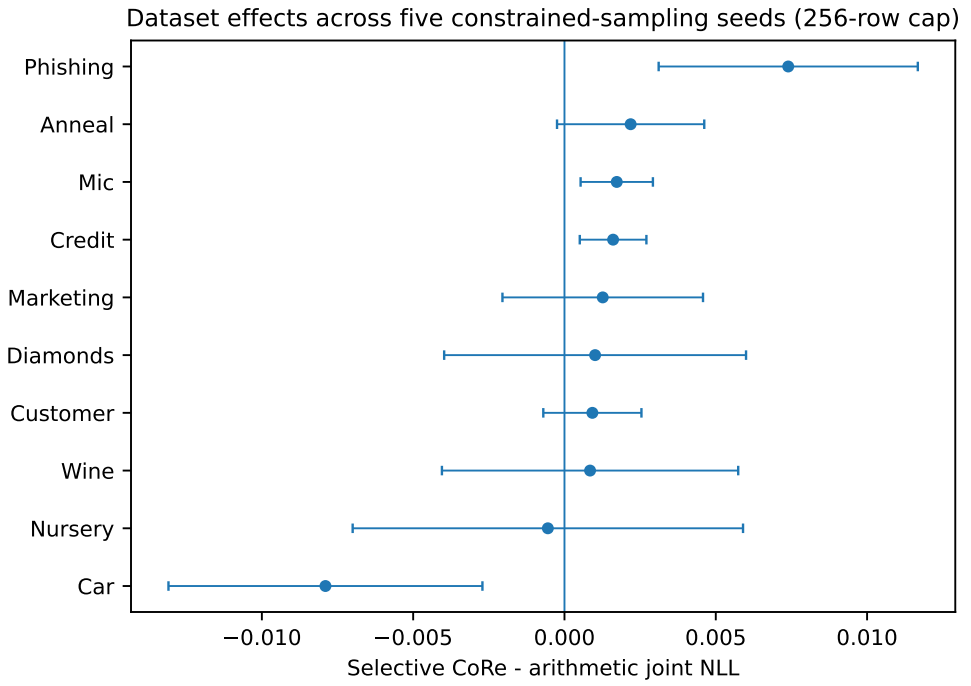


Figure 4: Dataset-level Selective-minus-arithmetic joint-NLL effects at the 256-row cap across five constrained-sampling seeds. Points show the seed mean and horizontal bars show one standard deviation across seeds. Negative values favor Selective CoRe.

Table 6
Selective-minus-arithmetic joint-NLL effect as the training cap and proportional validation pool increase. Positive values favor arithmetic pooling. The 256-row aggregate uses five seeds; other rows use three.

Train cap	Typical validation	Mean effect	TFM cells	Direction
64	13	+0.0065	60	Arithmetic better
128	26	+0.0025	60	Arithmetic better
256	52	+0.0009	100	Near parity
512	103	−0.0013	60	Selective better
1024	≈ 194	−0.0034	60	Selective better

Crucially, the relationship between inconsistency and Selective-CoRe benefit is not stable across models. In a frozen benchmark reliability analysis, the Pearson correlation between factorization TV and arithmetic-minus-selective gain is approximately −0.421 for TabICLv2 and +0.310 for TabPFN-3, with an overall correlation of only about −0.133 across all model cells. A high factorization TV therefore does not, by itself, tell us whether reconciliation will improve proper-score performance. The diagnostic must be interpreted together with model- and view-specific reliability.

This conclusion is also supported analytically: one can construct pairs of factorizations with large TV where one factorization is already equal to the truth and pooling necessarily hurts, as well as pairs with arbitrarily small disagreement symmetrically bracketing the truth where pooling helps. Inconsistency is a structural defect statistic, not an accuracy statistic.

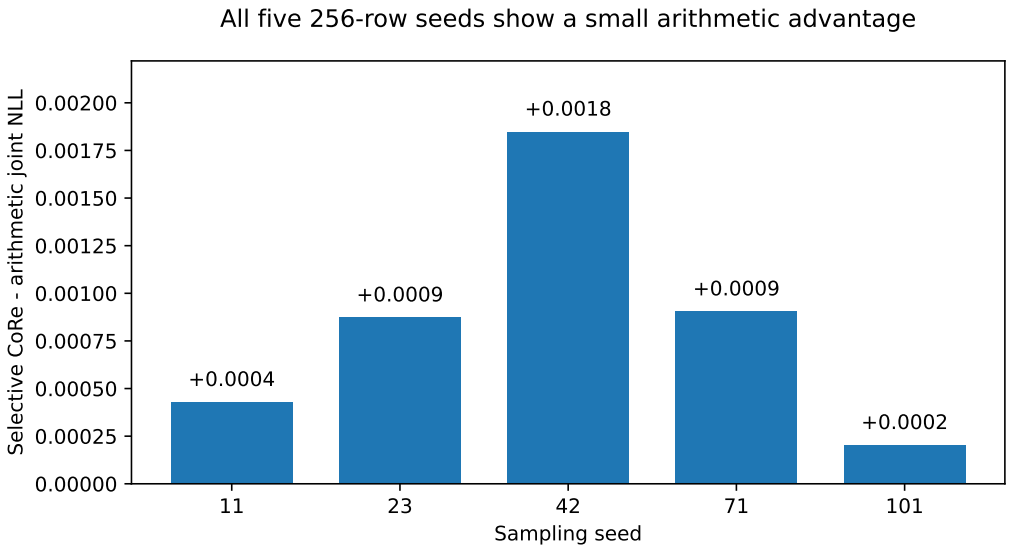


Figure 5: Seed-level Selective-minus-arithmetic NLL effects at the 256-row training cap. All five sampling seeds yield a small positive mean effect, but the magnitude varies materially across seeds.

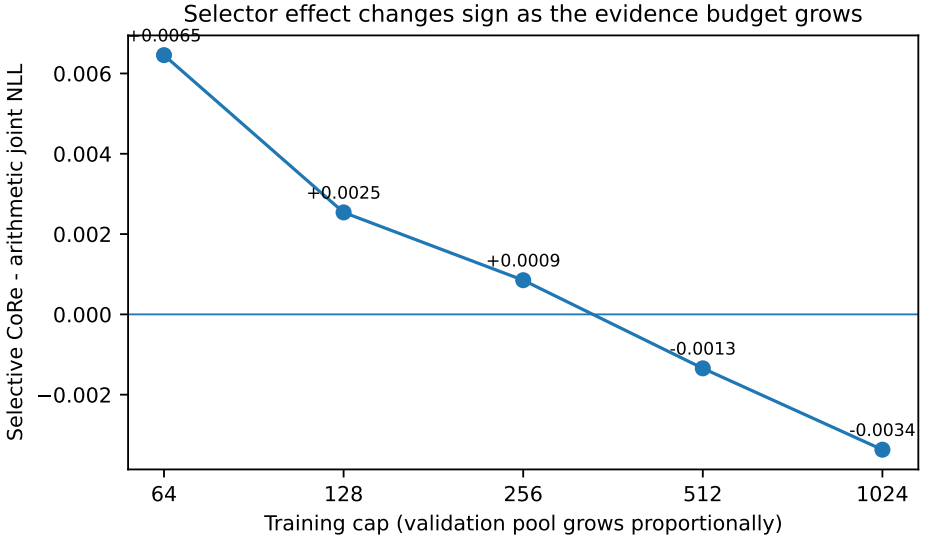


Figure 6: Evidence-budget transition. Positive values favor arithmetic pooling; negative values favor Selective CoRe. Training context and the held-out validation pool grow together, so this is an evidence-budget trend rather than a causal context-size effect.

6.7. Candidate-family complexity is measurable

The full 48-candidate selector is not uniformly superior to a smaller family. Table 8 reports mean joint NLL for nested candidate-family ablations aggregated over the robustness runs.

Pool-only selection is slightly better than the full family from 64 through 512 rows, while the full family becomes marginally best at 1024. This is consistent with a bias–variance trade-off in candidate selection: a larger family creates more oracle opportunity but also more ways to overfit a finite validation set.

Table 7
Model-specific mean Selective-minus-arithmetic NLL difference.

Model	64	128	256	512	1024
TabICLv2	+0.0120	+0.0056	+0.0044	−0.0004	−0.0027
TabPFN-3	+0.0009	−0.0005	−0.0026	−0.0023	−0.0041

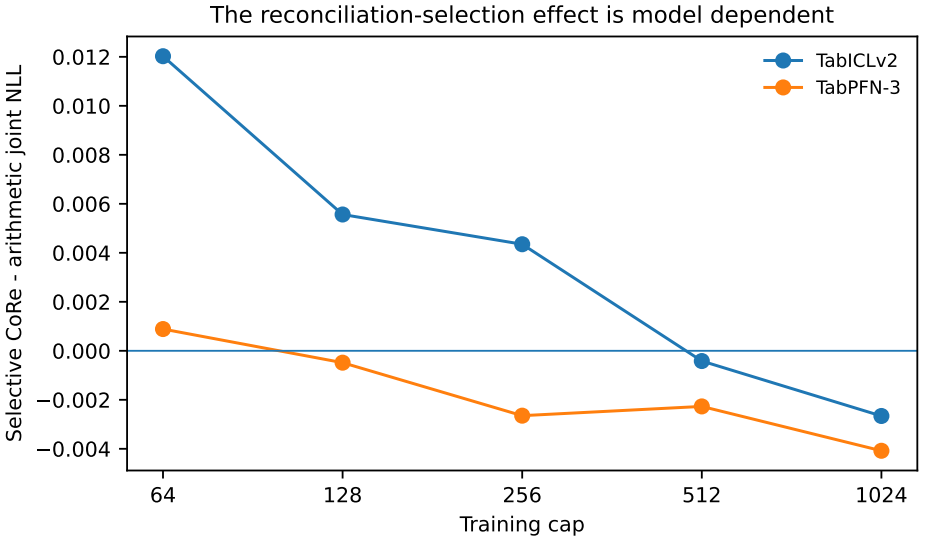


Figure 7: Model-specific evidence-budget effects. TabPFN-3 reaches a Selective-CoRe advantage earlier, while TabICLv2 pays a much larger small-budget selection cost and crosses the zero-effect line only at larger budgets.

Table 8
Mean joint NLL of validation-selected candidate families. Lower is better.

Train cap	Full	Pool only	Raw only	Repair only
64	1.6884	**1.6867**	1.6906	1.6885
128	1.4913	**1.4911**	1.4932	1.4912
256	1.3786	**1.3779**	1.3802	1.3788
512	1.2854	**1.2853**	1.2868	1.2854
1024	**1.2089**	1.2093	1.2103	1.2089

At the 256-row cap, a separate nested-validation sensitivity gives mean NLL 1.3805, 1.3798, and 1.3786 when using 25%, 50%, and 100% of the available validation pool, respectively. At the 64-row cap, using the full validation pool also improves mean NLL relative to the two smaller fractions. These ablations support the role of validation evidence while avoiding the stronger claim that every increase in validation size must improve every dataset.

6.8. Rare classes do not explain the 256-row effect

Some paired-target datasets contain very small target-class supports. We therefore perform a post-hoc exclusion sensitivity using data-set-level minimum target-class support. At the 256-row cap, keeping all ten datasets gives mean Selective-minus-arithmetic NLL +0.0009. Requiring minimum support of at least 2 leaves nine datasets and yields +0.0008; requiring at least 5 leaves six datasets and yields +0.0010; requiring at least 10 leaves five datasets and yields

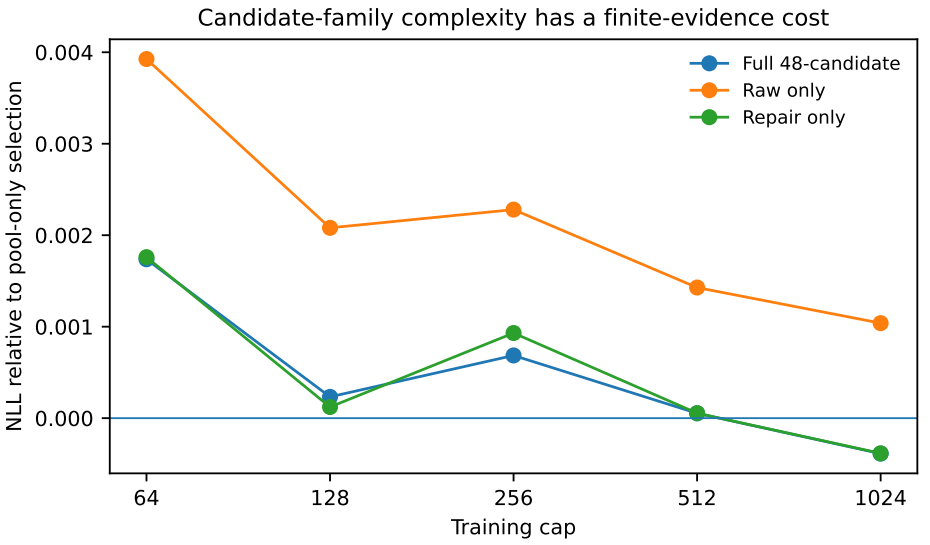


Figure 8: Candidate-family selection cost relative to the pool-only selector. Values above zero are worse than pool-only selection. The full 48-candidate family pays a measurable finite-evidence penalty at small and moderate budgets and becomes slightly better only at the largest budget.

+0.0007. None of the filtered analyses reverses the direction. Rare-class datasets therefore do not appear to be the sole explanation for the small 256-row disadvantage.

6.9. A conservative Safe-Selective rule is a negative control, not a win

The repository includes a Safe Selective CoRe rule based on structural-risk penalties. It falls back to arithmetic pooling unless a validation improvement exceeds the candidate-family complexity penalty. In 100 controlled known-truth tasks, the current rule selects the arithmetic fallback in all 100 cases, has zero mean difference from arithmetic, and mean oracle regret 0.0065. This establishes a useful negative control: a theoretically motivated guardrail can be so conservative that it eliminates both selection risk and selection benefit. We therefore do not present the current Safe-Selective rule as a performance contribution.

6.10. Ten-dataset fixed-test evaluation

We audited all 160 real-model cache/metadata pairs from the second protocol, checked 40 frozen split specifications, and recomputed each method's test scores directly from saved probability arrays. The primary comparison uses training cap 256, validation budget 52, four matched split settings per dataset, and ten dataset-blocked means per TFM. Table 9 gives mean joint NLL, with lower values indicating better proper-score performance.

The raw-or-pool selector shows the smallest mean NLL in both models, but the margin relative to fixed arithmetic pooling is only −0.000813 in TabICLv2 and −0.002212 in TabPFN-3. Among the 40 primary configurations per model, the restricted full selector chose hard or partial repair in 4 TabICLv2 cases and 8 TabPFN-3 cases. These are *selection events*, not evidence of repair benefit; averaged performance still does not beat the smaller raw-or-pool family. Dataset-blocked exploratory Wilcoxon p-values for these differences are 0.6953 and 0.2324, respectively, and their bootstrap intervals include zero. Hence the data do not establish reliable predictive superiority of the selector. Nor does the Protocol B full-family selection produce a demonstrable gain over the smaller raw-or-pool family. Notably, temperature fitting on a 52-example validation subset *worsens mean test NLL* under the current grid and splits; this is a finding about the implemented finite-label calibration procedure, not a general argument against calibration. The accompanying proper-score and decision metrics are displayed in Table 10; their aggregation preserves dataset balance and the distinction between log loss, squared-probability error, top-1 decisions and calibration diagnostics.

Table 9
Fixed-test benchmark: unweighted mean over ten datasets of joint NLL at training cap 256 and validation budget 52 (four paired split settings per dataset). *Full-family* denotes the fixed-test restricted eight-candidate selector, not the older 48-candidate method. Bold denotes the lowest descriptive mean in each model column, not a significant win.

Policy	TabICLv2	TabPFN-3
Arithmetic pool	1.34000	1.34147
Raw-or-pool selector	**1.33919**	**1.33926**
Full-family selector	1.33994	1.33944
Temperature-calibrated pool	1.35350	1.35492
Hard reconciliation	1.35831	1.35325
Halfway soft repair	1.34654	1.34524
Guarded calibrated repair	1.35350	1.35492

Table 10
Secondary proper-score and decision metrics in the new fixed-test protocol at training context 256, averaged equally across ten datasets (four split repeats within each dataset). NLL and multiclass Brier are lower-is-better, top-1 accuracy is higher-is-better, and ten-bin top-label ECE is an unstable descriptive diagnostic. Values are not significance tests.

TFM	Method	NLL ↓	Brier ↓	Acc. ↑	ECE ↓
TabICLv2	Arithmetic	1.3400	0.6148	0.4869	0.0669
TabICLv2	Raw-or-pool	1.3392	0.6144	0.4893	0.0671
TabICLv2	Restricted full selector	1.3399	0.6147	0.4877	0.0678
TabICLv2	Hard repair	1.3583	0.6180	0.4828	0.0679
TabICLv2	Halfway partial repair	1.3465	0.6157	0.4848	0.0693
TabICLv2	Temperature scaling	1.3535	0.6185	0.4869	0.0807
TabPFN-3	Arithmetic	1.3415	0.6142	0.4846	0.0555
TabPFN-3	Raw-or-pool	1.3393	0.6139	0.4824	0.0590
TabPFN-3	Restricted full selector	1.3394	0.6140	0.4828	0.0574
TabPFN-3	Hard repair	1.3533	0.6162	0.4848	0.0530
TabPFN-3	Halfway partial repair	1.3452	0.6147	0.4824	0.0544
TabPFN-3	Temperature scaling	1.3549	0.6176	0.4846	0.0712

6.11. Matched training-context sensitivity under fixed validation and test IDs

The second protocol holds each split's 205-example validation pool and 128-example test pool fixed while increasing the model's nested training context from 128 to 256 rows. This isolates a different factor from the earlier experiment in which upstream context and validation evidence changed together. Table 11 reports within-split paired changes in test NLL, with negative differences indicating improvement from the larger context. For both models, arithmetic pooling improves on 39 of 40 paired split configurations and on all ten dataset means. The corresponding mean decreases are 0.13332 (TabICLv2) and 0.11163 (TabPFN-3) NLL units. This reflects the benefit of additional task context under the recorded setup; it does not imply that a reconciliation policy is necessary or that the total label budget is limited to 256 examples.

6.12. Direct structural intervention: coherence is not accuracy

For each fixed test prediction, we apply the arithmetic-reference marginal projection from Eq. (9), holding the queried direct marginal distributions fixed. The marginal-discrepancy statistic is $\frac{1}{2}[\mathrm{TV}(Q_A, p_A) + \mathrm{TV}(Q_B, p_B)]$ averaged over test cases; it measures agreement *with the model's own direct predictions*, not distance to the unknown true marginal distribution. We independently measure whether this intervention decreases mean marginal total variation and whether the resulting held-out NLL improves. All 80 primary model–dataset–split configurations show reduced marginal discrepancy, but 67/80 show *increased* NLL. The reduction is expected from the construction of a marginal projection and should not be counted as an independent external-validation success; the held-out score comparison is the substantive empirical test. This isolates a substantive distinction: enforcing the selected marginal constraints changes the joint as intended, but the constraints may not be the most reliable information available about the true joint.

Table 11
Fixed-test, matched-context NLL differences, 256 minus 128 context rows. Each model has 40 matched split configurations (ten datasets, four repeats), but the reported descriptive paired test uses ten dataset-averaged differences. These results do not isolate model pretraining or architecture effects.

Model	Policy	Mean ΔNLL	Improved splits	Improved datasets
TabICLv2	Arithmetic pool	-0.13332	39/40	10/10
TabICLv2	Raw-or-pool	-0.13404	39/40	10/10
TabPFN-3	Arithmetic pool	-0.11163	39/40	10/10
TabPFN-3	Raw-or-pool	-0.10986	39/40	10/10

Table 12
Audited paired hard-reconciliation effects relative to arithmetic pooling. Positive ΔNLL/ΔBrier means worse predictive scoring. Bootstrap intervals and exploratory tests use ten dataset-level average effects; Holm correction covers the documented exploratory contrast family.

Model	Mean ΔNLL	Bootstrap 95% CI	Harmed	Wilcoxon p	Holm p
TabICLv2	+0.01831	[+0.00893, +0.02832]	9/10	0.00977	0.06836
TabPFN-3	+0.01178	[+0.00397, +0.01954]	9/10	0.01953	0.13672

The corresponding mean Brier increases are +0.00324 for TabICLv2 and +0.00199 for TabPFN-3. These bootstrap intervals exclude zero and the unadjusted paired p-values are below 0.05, yet both Holm-adjusted p-values exceed 0.05. Consequently we treat the widespread hard-repair penalty as a strong descriptive pattern with exploratory statistical support, rather than a family-wise-confirmed positive finding. These are exploratory per-model Holm adjustments over the reported candidate/reference contrast family; they do not exhaust all historical tests, budgets, models or additional post-hoc questions examined during research development. Bootstrap intervals excluding zero are likewise not evidence of universal protection against exploratory multiplicity. Partial (50%) soft repair reduces but does not eliminate the mean harm: its dataset-blocked ΔNLL is +0.00654 for TabICLv2 and +0.00377 for TabPFN-3. The intervention demonstrates a mechanism, but does not establish that all possible coherence-enforcing maps have the same effect.

6.13. Sensitivity to the projection reference distribution

The Protocol A hard projection and the Protocol B direct ablation use different KL-reference distributions: the Protocol A equal-weight variant projects the normalized geometric pool, whereas the fixed-test audit projects the arithmetic pool. To test whether the observed failure mode is an artifact of that choice, we performed an additional *post-hoc, inference-free* ablation using the frozen predictions from all 160 real-model configurations. The geometric-reference projection uses $\mathcal{P}_{p_A,p_B}(M_{0.5})$ with the same iterative proportional-fitting tolerances as the arithmetic-reference intervention. No checkpoint was reloaded, no hyperparameter was selected using test labels, and both projected joints were scored on the same saved held-out outcomes.

The geometric-reference projection also increases dataset-average NLL on nine of ten datasets for each model. Its arithmetic-pool-relative mean changes are +0.01800 for TabICLv2 and +0.01126 for TabPFN-3; both projections drive marginal discrepancy close to zero. The two reference distributions give numerically different repaired predictions, but their dataset-level patterns agree in this setting. The mean geometric-minus-arithmetic-reference repaired NLL changes are −0.00031 and −0.00052 respectively. These post-hoc results strengthen the *reference-sensitivity check* without making the historical penalty-based Soft CoRe or its 48-candidate selector equivalent to the restricted October analysis. No multiplicity-adjusted confirmatory claim is made for these new contrasts.

6.14. Dataset heterogeneity and the Diamonds exception

The sign of the repair effect is not universal. Across the ten datasets, hard projection harms mean NLL on nine datasets per model but improves it on Diamonds (mean ΔNLL −0.00513 for TabICLv2 and −0.01028 for TabPFN-3). Split-level analysis shows that TabICLv2's Diamonds improvement is driven by a large gain in the 71/0 split (−0.04562), while its other three splits are harmed. TabPFN-3 improves in three of the four Diamonds splits. Class-pair subgroup sizes can be sparse, so the observed advantage is an informative heterogeneous case, *not* proof of a broadly identifiable beneficial-repair regime. No dataset-specific rule was learned from test outcomes.

Table 13
Exploratory reference-distribution ablation at training context 256. Differences are dataset-balanced test NLL changes; positive values indicate harm. The last column counts datasets with worse geometric-reference repair than arithmetic pooling. The geometric reference corresponds to the equal-weight historical projection, not to the complete Protocol A 48-candidate selector.

Model	Hard(arith.) − arith.	Hard(geom.) − arith.	Hard(geom.) − geom.	Geom. repair harms
TabICLv2	+0.01831	+0.01800	+0.01836	9/10
TabPFN-3	+0.01178	+0.01126	+0.01192	9/10

6.15. Guarded repair and evidence-budget findings

The validation-guarded repair policy did not activate in its recorded runs and exactly matched its temperature-calibrated-pool fallback. It is therefore inappropriate to call the guard *effective at identifying beneficial repairs*; its observed behavior demonstrates nonactivation and an inability to outperform the fallback, not a safety guarantee under distribution shift. Likewise, the Protocol A 48-candidate selection results and the fixed-test eight-base-candidate results refer to different candidate sets and environment versions. Both experiments caution against treating a larger candidate family as intrinsically better when validation evidence is limited.

The two evidence-budget designs provide complementary, noninterchangeable evidence. In the historical protocol, training context and selection support increase together and the Selective-minus-arithmetic trend changes sign by the largest cap. In the audited fixed-test protocol, 128 versus 256 training contexts are nested within an unchanged 205-example validation pool and unchanged 128-example test pool, allowing context-size sensitivity to be studied independently of label availability for policy selection. Neither design constitutes a prospective deployment or independent-hospital evaluation, and neither is a confirmation that calibration or selection dominates all other post-processing strategies.

6.16. Targeted clean replication and provenance checks

A fresh-runtime check reproduced all six selected configurations at the level of the archived prediction arrays (Table 14). Both joint NLL and Brier values were unchanged at the displayed precision, and the maximum absolute elementwise probability difference was zero. These are six checks of previously evaluated configurations, not six additional independent samples. The recorded environment was Python 3.13.15, PyTorch 2.11.0+cu130, TabICL 2.2.0, TabPFN 9.2.0 and an NVIDIA Tesla T4. The newly configured TabICLv2 checkpoint-path SHA-256 was `bdc7dbd5...b3d4ed0`; the TabPFN-3 V3 checkpoint-path SHA-256 was `d0d865d5...18f3988`. Full fingerprints are retained in the run records. These observations confirm the identity of the configured checkpoint files in the new run, not the bytes actually loaded in all earlier experiments.

The initial online provenance run returned matching OpenML-reported MD5 checksums for ten of eleven raw source records, but the MIC metadata endpoint (OpenML ID 46980) returned HTTP 412. A subsequent standalone retrieval of the official MIC version 3 record succeeded (file ID 22125397): the downloaded 771,864-byte ARFF was reported to match the published MD5, and the required target attribute names were present. The follow-up ZIP retains the verification JSON, including its downloaded-file SHA-256, but not the raw ARFF bytes; therefore we treat this last match as a documented external verification rather than an independently repeated download in this audit. The two reports together account for eleven matched raw-source checks. Neither raw-file checksums nor the six selected replication runs establish byte-identical preprocessing or checkpoint provenance for all original GPU executions.

7. Discussion

The difficulty with probability reconciliation is not finding a distribution that obeys the marginal constraints. For positive categorical probability tables, iterative scaling provides a direct way to do that. The difficult part is deciding whether the constraints are worth imposing. Separate TFM queries can disagree because their approximations are imperfect, but the directly queried marginals are also approximations. A projection treats those marginals as authoritative, even when they are less reliable than one of the chain-rule forecasts.

The controlled studies make this dependence explicit. When the supplied marginals are close to the known truth, projection can improve the expected log score; when they are corrupted, the same operation transmits their errors

Table 14
Independent six-job clean replication, each with context 256, seed 23 and split index 0. Archived and replicated arithmetic-pooling NLL are identical; maximum absolute difference over saved probability tensors is 0 for every row.

Dataset	Model	Archived NLL	Replicated NLL
Car	TabICLv2	1.319614	1.319614
Car	TabPFN-3	1.322587	1.322587
Nursery	TabICLv2	1.348436	1.348436
Nursery	TabPFN-3	1.375204	1.375204
Diamonds	TabICLv2	2.561353	2.561353
Diamonds	TabPFN-3	2.534167	2.534167

to the joint. In the fixed-test benchmark, we cannot measure the true marginal distribution, but we can measure the consequences for held-out outcomes. Arithmetic-reference hard projection reduced marginal mismatch in all 80 primary comparisons at the 256-row context and increased NLL in 67. The geometric-reference sensitivity analysis showed a similar pattern. The observation is therefore not confined to one pooling reference, although it should not be generalized to every possible reconciliation objective.

It would also be misleading to describe repair as uniformly harmful. Diamonds had a favorable mean repair effect under both models, driven in part by specific split realizations. These cases deserve further study because a useful reconciliation rule must recognize them without making larger mistakes elsewhere. Our validation-guarded rule did not meet that test: it chose its calibrated-pooling fallback on every evaluated occasion. This prevented repair-induced changes in those cases, but does not establish that the gate can detect a beneficial intervention.

The simple raw-view-or-pool selector offers a different lesson. It was marginally better than arithmetic pooling on average in the ten-dataset fixed-test study, but its advantage was small and was not supported by the adjusted exploratory tests. Increasing the candidate family to include more reconciliation variants produced no consistent additional benefit. This is compatible with ordinary model-selection variance: every extra candidate creates another opportunity for a small validation set to favor a prediction that does not generalize (Cawley and Talbot, 2010). In the five-seed analysis, the gap between oracle candidate performance and attainable validation selection was substantial, especially for TabICLv2. A post-hoc method should therefore be evaluated as a selection procedure, not only by the quality of its best possible output.

We deliberately report two real-model protocols separately. Protocol A varies both training context and available selection evidence, so its budget-dependent sign changes cannot isolate the reason for an apparent gain. Protocol B holds validation and test IDs fixed while changing the training context. It shows that larger contexts improve pooled forecasts in nearly every paired split, but that is a property of the underlying model and training information rather than of reconciliation. Combining the protocols into a single inferential sample would obscure these distinctions.

The practical implication is limited but concrete. An inconsistency score can tell us that the four queried probability views are not describing the same joint distribution. It cannot, on its own, tell us which view to believe. Proper scores on genuinely held-out outcomes, classwise support, and stability across samples are more direct evidence for deciding whether to select, recalibrate, or repair. Developing a gate that makes this decision reliably across target structures and model families remains open.

Finally, artifact checks and fresh inference test different aspects of reproducibility. The audit verifies frozen split IDs, cache bytes, normalized probabilities, and the arithmetic of the reported scores. Six fresh GPU executions reproduced the corresponding archived prediction arrays exactly, while recording configured checkpoint-file fingerprints. Those records give strong evidence for the six selected configurations, but do not retrospectively establish which checkpoint bytes were loaded in every benchmark job. The later MIC verification report closes the raw-download checksum gap, but does not independently establish the historical feature transformation and row-selection pipeline. Reporting this boundary is preferable to treating source-file authentication as complete end-to-end reproducibility.

8. Limitations

The formulation is pairwise and categorical. Global compatibility among three or more targets can impose constraints that are not visible in pairwise reconciliation, and mixed continuous targets require different numerical treatment. Extending CoRe-TFM to graphical multi-target reconciliation is future work.

The primary real-model claim covers only two released TFMs. CatBoost is intentionally treated as a boundary baseline rather than a third TFM. TabFM support exists in the codebase, but the frozen protocol excluded it before comparative outcome inspection for recorded operational-compatibility and runtime reasons. Because the archive contains neither completed TabFM folds nor a controlled TabFM timing study, the paper makes no claim about TabFM accuracy or intrinsic computational efficiency. Broad statements about all TFMs would therefore be unsupported, and replication with additional released TFMs remains a priority.

The datasets are small to medium tabular benchmarks and the protocol is explicitly bounded. The 1024-row cap is not attainable for every dataset, causing the effective training/validation counts to vary at the largest budget. The context experiment also couples upstream training context with validation size, preventing causal separation of the two effects.

The conditioning procedure fits separate estimators for each of four target views and enumerates hypothetical category values at inference. Conditioning on enumerated outcome categories can create feature configurations that are uncommon within the training context, and conditional probabilities from separate fits need not be mutually compatible. The training-pool class-coverage frontloading also creates deliberately non-random nested training subsets, which can favor coverage over representative sampling. These features belong to the experimental protocol rather than being neutral implementation details. It consequently incurs extra fitting and prediction cost relative to a single target classifier. Cached wall-clock values include possible checkpoint loading and download, so they are not valid steady-state throughput comparisons. No full latency, memory, energy or production-cost benchmark is claimed here.

Only ten datasets define the primary inferential unit. The context-size rows above 256 use three seeds, and the numerical sign transition should therefore be treated as descriptive robustness evidence, not as a powered proof of monotonic improvement. The multiple exploratory analyses in this paper are used for mechanism discovery; the primary confirmatory interpretation remains dataset-blocked and focused on proper scoring rules.

Rare classes remain challenging despite the exclusion sensitivity. Post-hoc support filtering cannot replace a purpose-built rare-event benchmark. Likewise, ECE is binning dependent and is not used as the main inferential endpoint. The controlled reliability mixture is a designed stress test, not an estimate of how frequently different reliability regimes occur in deployed applications.

The candidate-family oracle is non-deployable because it uses test outcomes and is reported only to quantify headroom. Safe Selective CoRe currently collapses to the arithmetic fallback and therefore requires redesign before it can be proposed as an improvement.

Limits of the matched fixed-test evaluation. The second study is based on saved GPU prediction caches with an independent recomputation of their losses, split IDs and probability constraints. Six subsequent clean-runtime executions reproduced selected cache arrays exactly and recorded replication model-file fingerprints, but cannot authenticate the loaded checkpoint bytes in all 160 historical executions. The initial online run rehashed ten of eleven raw OpenML source records; a subsequent standalone MIC version 3 download reports matching official MD5 and records its SHA-256, completing the documented eleven-source checksum matrix. End-to-end identity of historical processed features and row transformations still requires a cross-protocol schema audit, particularly for Diamonds. Five processed pickle snapshots in the earlier extension archive are not substitutes for raw dataset records. The two real-model protocols differ in package versions, split construction and candidate sets, and their inference is never pooled. The validation reservation of 205 labels alongside a training context as low as 128 is a deliberately decoupled diagnostic, not a fixed-total-label setting. Repeated splits are correlated within each dataset; uncertainty is therefore computed across ten dataset-level effects. After exploratory multiple-comparison adjustment, the hard-repair result does not meet the conventional 0.05 threshold. The geometric-reference bridge was motivated after inspecting earlier findings, making it exploratory. Neither stable throughput nor deployment impact was established.

9. Reproducibility

The implementation is organized into model adapters, probability extraction, proper-score metrics, reconciliation modules, benchmark auditing, robustness runners, and reliability-analysis utilities. Unit tests cover reconciliation, soft scaling, selective policies, metrics, synthetic truth, benchmark auditing, robustness execution, and research extensions.

The five-seed benchmark records 1,200 method rows over 150 model–fold tasks, environment versions, fold manifests and checksums. Such provenance is in the spirit of documented machine-learning benchmarking infrastructures such as OpenML (Vanschoren et al., 2014). Supplementary Material S1 extends this evidence to five 256-row seeds and twelve additional context-budget runs (three seeds at each of four other caps), plus selection-family ablations, validation-fraction

studies, reliability summaries, rare-class analyses and CPU-only controls. Across all 17 runs it records 20,400 method rows. Dataset–model–fold–seed cells appear eight times, once for each summarized method; those rows are not independent observations for significance tests.

The robustness preflight records Python 3.13.15, PyTorch 2.11.0 with CUDA 12.8, TabICL 2.1.1, TabPFN 8.4.0, CatBoost 1.2.10, and an NVIDIA Tesla T4 environment for the archived execution. The benchmark source commit is pinned independently of later manuscript and analysis revisions.

The code is available at `https://github.com/bnssaanirudh/core-tfm`. The robustness claims were checked against the separately archived *Supplementary Material S1*, rather than inferred from the public repository status summary. S1 includes the complete 17-run aggregate, individual fold-result tables, run-completion records, ablations and environment details. Its 169 listed file SHA-256 values and sizes match the corresponding archived files, and recomputation of the ten dataset effects at the 256-row cap reproduces +0.0008510181 and $p = 0.10546875$. This checks archive integrity and the arithmetic of stored results; it does not constitute a fresh rerun of model inference. For a reproducible submission, the final supplementary archive should be permanently linked to an immutable public source revision or repository release.

Matched fixed-test evidence and independent audit. The matched-context study provides two complementary archived objects: a combined evidence ZIP retaining 160 original per-configuration NPZ/JSON prediction caches, 40 frozen split descriptions and associated summaries, and an independent numerical-audit ZIP containing directly reconstructed policy scores, paired intervention effects, dataset-blocked uncertainty, figures, integrity records and environment/provenance-gap reports. The combined manifest lists 414 verified source entries, while the audit bundle lists 23 verified output artifacts. The saved-output audit reproduces 1,920 method-level evaluation rows. The supplementary two-model audit preflight records TabICL 2.2.0, TabPFN package 9.2.0, PyTorch 2.11.0+cu130 and Tesla T4 hardware (these are not the older benchmark's versions). Hash and score agreement establish artifact integrity and arithmetic reproducibility, *not* that the exact original weights or raw dataset bytes were independently identified. Both archives should be published immutably with clear provenance boundaries and without sensitive authentication material. The independent-audit procedure does not unpickle untrusted processed-data files.

Clean replication and raw-data verification. A six-job GPU replication re-evaluated Car, Nursery and Diamonds under the same saved sample IDs and 256-row contexts, with both primary models. The six probability arrays were identical to the stored historical arrays, and the reconstruction of test NLL and Brier produced zero differences at numerical precision. The new execution report includes explicit configured checkpoint-file SHA-256 values, package versions, CUDA hardware and per-job status. These replication records are not backfilled into the older model metadata. An initial online OpenML provenance check matched reported MD5 checksums for ten of eleven raw source records. The remaining MIC record (OpenML 46980) was subsequently retrieved in a standalone check after an earlier HTTP 412 response; its official version 3 metadata lists file ID 22125397, and the separate verification report records a downloaded-file MD5 match and SHA-256 fingerprint. The report is a provenance record, not the raw dataset itself; the older experiment's exact preprocessing identity is not inferred from it. No frozen analysis result was altered. Six clean replications have not changed the original $n = 160$ experimental count.

10. Conclusion

Four probability queries about the same pair of outcomes need not agree when answered separately by a tabular foundation model. CoRe-TFM examines whether enforcing agreement also improves forecasting. The controlled experiments show that the answer depends on the reliability of the marginals used as constraints; the real-model results show why that distinction matters.

Across ten paired-target datasets and two released TFMs, hard projection reduced marginal inconsistency in all 80 primary fixed-test comparisons, yet increased held-out NLL in 67. The dataset-balanced mean NLL increases were +0.01831 for TabICLv2 and +0.01178 for TabPFN-3. The adjusted exploratory tests do not justify a universal claim of harm. Diamonds supplied counterexamples, and the broader reconciliation candidate family did not establish an advantage over simpler selection and pooling baselines. A guarded repair policy did not activate, so its potential benefit remains untested.

These observations suggest a straightforward principle for systems that use overlapping predictive probabilities: measure structural consistency and predictive quality separately. A consistent joint is mathematically attractive, but

predictive trust must come from outcomes and reproducible evidence, not consistency alone. Future work should concentrate on learning when a constraint is credible, extending the analysis beyond two categorical targets, and testing such decisions under independent dataset and deployment shifts.

A. Rare-class sensitivity analysis

B. Dataset-level comparisons

The following paired effects are computed from the archived 256-row real-model results after averaging over folds, both primary TFMs and the five seeds within each dataset. The test uses these ten dataset effects, not the 500 underlying TFM model–fold–seed cells.

C. Replication and source-data provenance

The clean replication took place after the fixed-test benchmark and was not used to choose a predictive policy. The software environment was Python 3.13.15, PyTorch 2.11.0+cu130, TabICL 2.2.0, TabPFN 9.2.0, NumPy 2.1.3, pandas 2.2.3, SciPy 1.16.3 and scikit-learn 1.6.1 on an NVIDIA Tesla T4. The replication model-file SHA-256 fingerprints were:

```
TabICLv2: bdc7dbd5e4ff21f8f0456fcf90c6b7cdf72dbea960f2d05b19bec19f9b3d4ed0
TabPFN-3: d0d865d54dfbc524f5703104be90620182dca7e5fb2c16de72e9959ea18f3988
```

For TabICLv2, the configured checkpoint repository revision was `4dcd344ece2c00be9e831fdd35bed57b5ad83e19`. For TabPFN-3, the explicit version was V3. These identifiers describe the fresh run only. The archived historical cache sidecars continue to record an unverified checkpoint-weight SHA-256.

The initial online provenance archive retains SHA-256 values and metadata for its ten matched source records. A subsequent standalone MIC verification JSON separately records the official and downloaded MD5 values `4af33fbb77eee0c5600aaa424efba5bc`, raw-file SHA-256 `402ab2d964ed2c23da168e1a6d264f76926fc9365191f492b32b7bbe3c7f08fe`, file ID 22125397, and an ARFF length of 771,864 bytes. The downloaded ARFF was not included in the small verification ZIP; consequently its byte-level digest has not been independently recomputed during manuscript preparation. The six clean-run execution events remain in their original report. We do not infer that every historical processed input or checkpoint was byte-identical to the newly verified sources.

Data and code availability

Code is available at `https://github.com/bnssaanirudh/core-tfm`. The historical five-seed experimental evidence (S1) and the separate 160-configuration prediction archive and numerical audit (S2) are distinct evidence streams and should remain separately labeled. A subsequent clean replication report records six new runs and checkpoint-file fingerprints; ten raw OpenML records passed its online checks, and a separately archived official-source verification later reported a matching MD5 for MIC version 3, completing the eleven-record verification matrix. An unpublished, checksum-verified S2 deposit candidate has been prepared, but a permanent public DOI has not yet been issued. The data availability statement must be updated with the immutable source release, archive DOI and license terms before submission. Model weights, credentials and raw datasets are not redistributed in the candidate deposit. Results in the article can be read without a separate supplementary PDF; the machine-readable prediction caches and verification scripts remain external research artifacts.

CRediT authorship contribution statement

Badampudi Agasthya Anirudh: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft, Writing – review & editing. **Harshita Bogineni:** Methodology, Investigation, Validation, Data curation, Formal analysis, Writing – review & editing. **Sri Preethaa K R:** Supervision, Methodology, Project administration, Validation, Writing – review & editing.

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Table 15
Rare-class exclusion sensitivity at the 256-row cap. Positive effects mean Selective CoRe is worse than arithmetic pooling.

Minimum support	Datasets kept	Mean effect	Wins	Wilcoxon p
1	10	+0.0009	2	0.105
2	9	+0.0008	2	0.164
5	6	+0.0010	1	0.438
10	5	+0.0007	1	0.625

Table 16
Five-seed dataset-level Selective-CoRe minus arithmetic joint NLL at the 256-row training cap. Negative values favor Selective CoRe. Values are shown with additional precision to facilitate an independent reconstruction of the paired statistic.

Dataset	Mean paired difference
Car	−0.007898733
Nursery	−0.000549423
Wine	+0.000845628
Customer	+0.000921804
Diamonds	+0.001012085
Marketing	+0.001264137
Credit	+0.001606358
MIC	+0.001727603
Anneal	+0.002186410
Phishing	+0.007394310
Dataset mean	+0.000851018

Table 17
Raw OpenML source provenance outcomes in the independent online check. Wine draws on two source IDs, giving eleven records for ten paired datasets.

Paired dataset	OpenML ID(s)	Raw-record verification
Anneal	46906	MD5 matched
Car	40975	MD5 matched
Credit	46918	MD5 matched
Customer	46938	MD5 matched
Diamonds	42225	MD5 matched
Marketing	46940	MD5 matched
MIC	46980 (v3)	MD5 matched in follow-up check
Nursery	26	MD5 matched
Phishing	46963	MD5 matched
Wine	40691, 40498	Both MD5 values matched

Declaration of competing interest

The authors declare no competing interests.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

The authors used ChatGPT for language editing and manuscript organization. All analyses, sources and conclusions were reviewed by the authors, who take responsibility for the content.

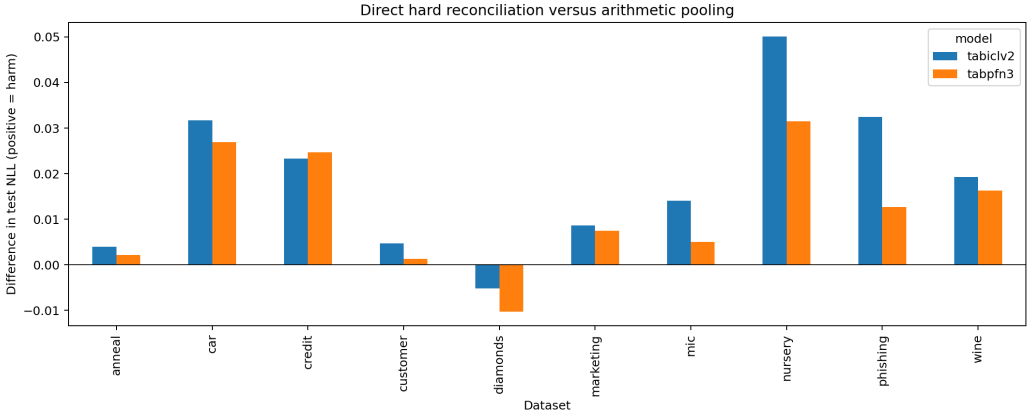
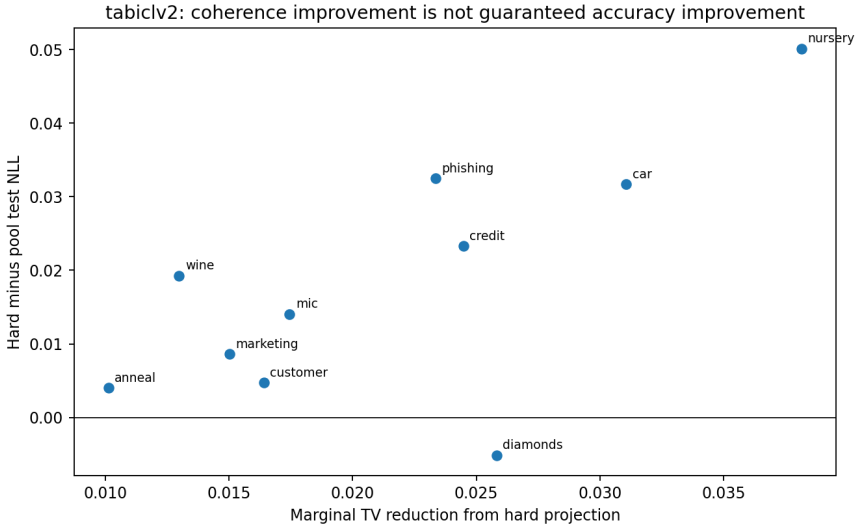
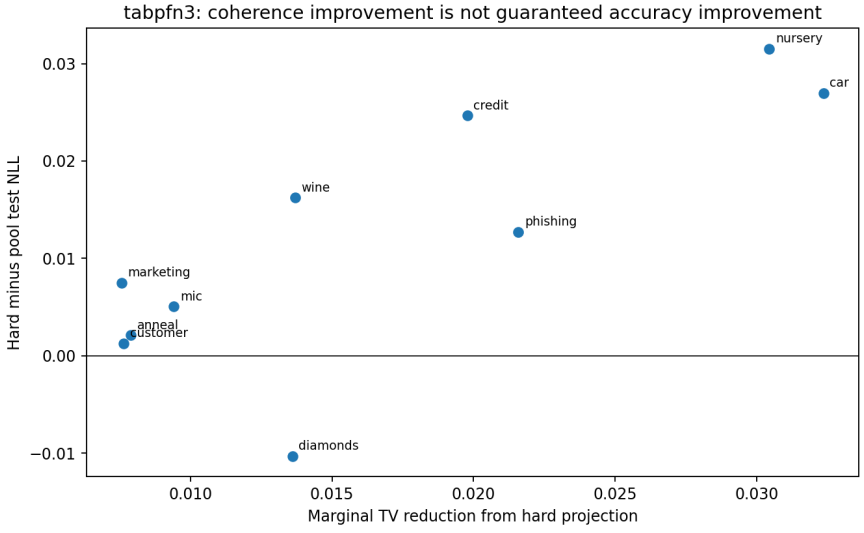


Figure 9: Independent fixed-test audit of hard-reconciliation NLL change by dataset and model at training cap 256. Positive differences indicate degraded predictive log loss after repair. Each dataset value is averaged across four randomized split settings.



(a) TabICLv2



(b) TabPFN-3

Figure 10: Consistency–accuracy diagnostics from the audited 160-cache study. The two models are shown separately because the relationship between changes in marginal discrepancy and test NLL depends on the model and dataset; better structural agreement is not a monotone predictor of improved log loss.

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